

[DC] Augmented Co-Embodiment for Motor Skill Learning with Held Tools in Virtual Reality

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ABSTRACT

Virtual reality (VR) is an interesting tool for developing new pedagogical methods for motor skill learning, enabling to give more varied visual concurrent feedback. Co-embodiment allows to share the control of an avatar between two persons. This technique has become popular in virtual reality for studying the relationship between the sense of agency and embodiment of the co-controlled avatar. Recently, some studies have begun to propose motor skill learning methods based on virtual co-embodiment. I propose to improve and extend these methods on three main topics. (Q1) Investigate whether additional concurrent feedbacks alongside virtual co-embodiment could improve learning performances. (Q2) Extend the results of embodiment and co-embodiment to an external held object, such as a crafting tool. (Q3) Propose more efficient co-embodiment methods for motor skill learning, without mobilising a human instructor. Our objective is to improve motor skill learning performance in craftsmanship, especially when using held tools, paying particular attention to the conscious and unconscious learning processes.

Index Terms: Virtual Reality, Virtual co-embodiment, Motor skill learning.

1 BACKGROUND AND MOTIVATION

Learning motor skills is an important part of many activities, and can be relevant at different ages, particularly for medical rehabilitation, or when it comes to learning craft skill. A motor skill can be described by 3 main characteristics:

1. optimal movement selection and execution (A)
2. improved movement speed and accuracy (B)
3. reduced movement variability and error (B)

where

- (A) represents conscious / strategic motor control
- (B) represents unconscious / non-strategic motor control [16]

Virtual reality (VR) is quite a powerful tool when it comes to learning gestures. It improves the sense of agency, the sense of embodiment, as well as precision [14, 7, 5]. Thanks to these qualities, VR makes learners more active in their training, improving the autonomy and the self-confidence [14]. VR can provide more original visual concurrent feedback during practice that would be very difficult to provide precisely in the real world. It is also possible to create new movement visualisations for the learner, such as seeing the expert perform the movement in a 1st person point of view (ghost movement) [4]. VR also allows to decompose the gestures for the learner, both in time and space, ensuring better understanding and facilitating movement integration [6, 15].

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Another line of work is the use of co-embodiment to teach motor skills in virtual reality [5, 13]. In this approach, the learner shares the controls of an avatar with the instructor (who maybe human or an agent), each weighting usually for 50% of the avatar's movement [2]. This results in a visuo-proprioceptive offset that influences the learner's motion [1, 3]. However, the effects of co-embodiment on skill retention have only begun to be investigated.

In my thesis, I aim to further investigate and improve learning motor skills through co-embodiment. In particular, I would like to study how it can affect the conscious and unconscious skill acquisition, to provide better tools for learning motor skills. As my thesis also focuses on craft know-how, I intend to try to extend the results of co-embodiment learning to in-hand tools, which are commonly used for such activities. To this end, I ask the following questions:

- **Q1:** Can additional error feedback improve co-embodiment motor skill learning?
- **Q2:** How can we embody an external held tool and extend co-embodiment results to the manipulation of this tool?
- **Q3:** Is it possible to create more efficient co-embodiment methods for motor skill learning, without mobilising a human instructor?

2 CO-EMBODIMENT AND EXTERNAL ERROR FEEDBACK FOR MOTOR SKILL LEARNING (Q1)

Feedback is primordial in learning. In classic motor skill learning, it is difficult to provide efficient and easily understandable concurrent feedback [4]. Virtual reality offers this advantage, as the learning environment can be fully controlled, both in terms of displaying information and choosing the learning scenario [8].

While co-embodiment itself is a concurrent feedback, I would like to explore whether co-embodiment can be augmented to learn more effectively. Actually, current studies usually provide additional feedback alongside co-embodiment, such as visual boolean feedback on failure or success [5, 13]. It does not seem clear how this additional feedback contributes to the co-embodiment learning results. Therefore, we would define, for this experiment, a baseline co-embodiment without additional error indicators, with 50% of ownership for both learner and expert. Using this baseline, the idea would be to see whether additional feedback optimizes the learning performances: boolean feedback on success/failure, gradual feedback on the error size, direction error feedback and surimpression of the expert's movement. Through this experiment, I would also like to assess self attribution of co-embodied movement. Indeed, it has been shown that one tends to self-attribute distorted movements as long as they are helpful [3].

In the definition of skill we chose, conscious and unconscious motor skill control have distinct characteristics. It would be then interesting to consider how to improve learning skills for both conscious and unconscious learning processes, in order to facilitate the right type of feedbacks depending on the needs of the learner or the task to accomplish. My hypothesis is that the fewer additional

feedback information, the more unconscious learning would be improved, as the learning process would get closer to implicit learning methods [11]. To evaluate this hypothesis, we plan to categorize conscious and unconscious effects of the learning feedbacks, using adapted metrics. The conscious motor processes would be characterised by measuring the movement selection and its successful execution. For the unconscious motor processes, it would be by measuring the variability, velocity and accuracy over several movement repetitions.

3 CO-EMBODIMENT OF EXTERNAL IN-HAND TOOL (Q2)

Craftsmanship often involves using in-hand tools for crafting. Manipulating them is necessary for mastering and transferring the skill to real-world applications. Virtual reality allows objects, such as crafting tools, to be easily tracked and rendered in a virtual environment. It has been shown that users can extend their sense of embodiment to external objects when they are held in hand, both in virtual reality and in the real world [12].

Though, for co-embodiment learning, we must first extend the results to a co-manipulated external object for its embodiment qualities and its learning performance. The idea would be to find a minimal and optimal *body and tool* representation that ensures embodiment in this process. If this were not a full body representation, it would not be necessary to use inverse kinematics (IK) in co-embodiment, which may lead to inaccurate or uncanny body representations, detrimental to learning.

4 TOWARDS BETTER DESIGN OF CO-EMBODIMENT AGENTS FOR MOTOR SKILL LEARNING (Q3)

Learning with human co-embodiment requires finding a teacher who masters both the movements to teach and how co-embodiment works in virtual reality [5]. One solution to this issue is to replace the human instructor with an agent [13]. Using an agent could also potentially offer more control over the feedback evolution that is given to the learner, and reduce the performance drop after training that can be an issue in co-embodiment training [13].

Co-embodiment for learning motor skill imitates a common practise among instructors: applying forces to the learner's body or tool to guide it towards the ideal movement. In most co-embodiment methods, the force is modelised by a barycenter method between the teacher's and learner's positions at every time, attracting the learner towards the good position [5, 13]. In a fixed weight co-embodiment setup, there is still room for improvement in co-embodiment learning methods, by conceiving and implementing new ways of presenting concurrent actions. For example, it is possible to use the displacement vector for co-embodiment instead of the position barycentre [9]. Furthermore, the displacement vector method could be extended by using a larger past window to provide smoother feedback. Alternatively, we could also propose non-hand-position-driven approaches for co-embodied feedback.

Finally, learning performance usually improves when help is gradually decreased and sometimes even removed during the learning process, giving the learner a better assessment of their learning progression [10]. For now, static weights of around 50% have been used in co-embodiment experiments [13]. I would like to explore using dynamic weights that match the learner progression during his training, as a means of improving learners' performance.

5 CONCLUSION

The proposed topics for my thesis are aimed at advancing pedagogical methods for motor skill learning in virtual reality, and the co-embodiment of held objects. With a focus on learning metrics,

my line of work is to study how co-embodiment could stimulate conscious and unconscious learning processes.

As we are already planning experiments on Q1, I would most likely like to discuss Q2 and Q3, to see how I can develop the ideas I got from the literature and make them fit the overall story. I would also be glad to discuss metrics for both learning motor skills and the concept of co-embodiment of held tools.

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